31) c) $\forall x \exists y (P(x,y) \wedge \exists z R(x,y,z))$

$\neg \forall x \exists y (P(x,y) \wedge \exists z R(x,y,z)) \equiv \exists x \neg \exists y (P(x,y) \wedge \exists z R(x,y,z))$ (De Morgan's law of quantifiers) $\equiv \exists x \forall y \neg (P(x,y) \wedge \exists z R(x,y,z))$ (De Morgan's law of quantifiers) $\equiv \exists x \forall y (\neg P(x,y) \vee \neg \exists z R(x,y,z))$ (1st De Morgan's Law) $\equiv \exists x \forall y (\neg P(x,y) \vee \forall z \neg R(x,y,z))$ (De Morgan's law of quantifiers)

d) $\forall x \exists y (P(x,y) \longrightarrow Q(x,y))$

$\neg \forall x \exists y (P(x,y) \longrightarrow Q(x,y)) \equiv \exists x \neg \exists y (P(x,y) \longrightarrow Q(x,y))$ (De Morgan's law of quantifiers) $\equiv \exists x \forall y \neg (P(x,y) \longleftrightarrow Q(x,y))$ (De Morgan's law of quantifiers) $\equiv \exists x \forall y \neg (\neg P(x,y) \vee Q(x,y))$ (conditional-disjunction equivalence) $\equiv \exists x \forall y (\neg(\neg P(x,y)) \wedge \neg Q(x,y))$ (2nd De Morgan's Law)

$\equiv \exists x \forall y (P(x,y) \wedge \neg Q(x,y))$ (Double negation Law)

32) Express the negations of each of these statements so that all negation symbols immediately precede predicates.

a) $\exists z \forall y \forall x T(x,y,z)$

$\neg \exists z \forall y \forall x T(x,y,z) \equiv \forall z \neg \forall y \forall x T(x,y,z)$ (De Morgan's law of quantifiers)

$\equiv \forall z \exists y \neg \forall x T(x,y,z)$ (De Morgan's law of quantifiers)

$\equiv \forall z \exists y \exists x \neg T(x,y,z)$ (De Morgan's law of quantifiers)

b) $\exists x \exists y P(x,y) \wedge \forall x \forall y Q(x,y)$

$\neg (\exists x \exists y P(x,y) \wedge \forall x \forall y Q(x,y)) \equiv \neg \exists x \exists y P(x,y) \vee \neg \forall x \forall y Q(x,y)$ (1st De Morgan's Law)

$\equiv \forall x \neg \exists y P(x,y) \vee \exists x \neg \forall y Q(x,y)$ (2 applications of De Morgan's law of quantifiers)

$\equiv \forall x \forall y \neg P(x,y) \vee \exists x \exists y \neg Q(x,y)$ (2 applications of De Morgan's law of quantifiers)

c) $\exists x \exists y (Q(x,y) \longleftrightarrow Q(y,x))$

$\neg \exists x \exists y (Q(x,y) \longleftrightarrow Q(y,x)) \equiv \forall x \neg \exists y (Q(x,y) \longleftrightarrow Q(y,x))$ (De Morgan's law of quantifiers) $\equiv \forall x \forall y \neg (Q(x,y) \longleftrightarrow Q(y,x))$ (De Morgan's law of quantifiers)